TURBOCHARGER WITH VARIABLE TURBINE GEOMETRY [SKYACTIV-D 1.5]

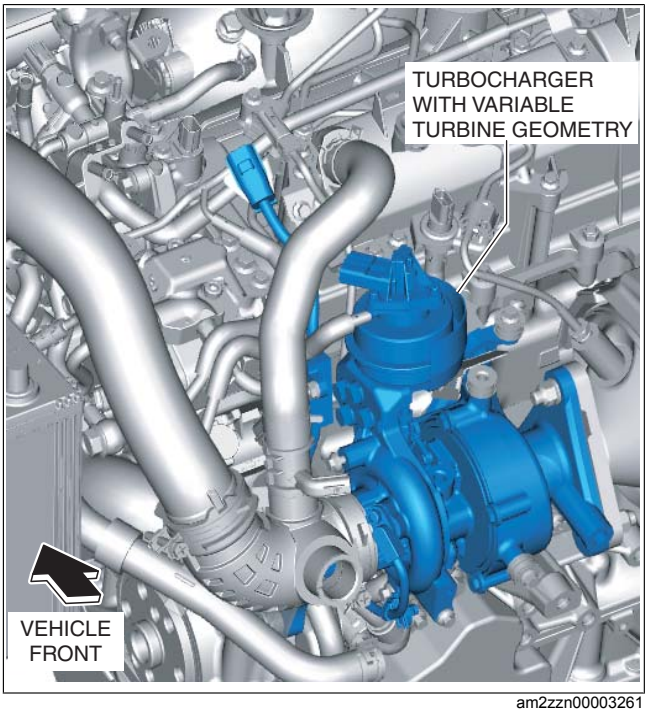
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Purpose, Function

- The turbocharger with variable turbine geometry uses the pressure of the exhaust gas to pressurize the intake air and supply it to the cylinders.
- A turbocharger with variable turbine geometry equipped with variable nozzles in the turbocharger has been adopted to improve the response at all speed ranges from low to high vehicle speeds.

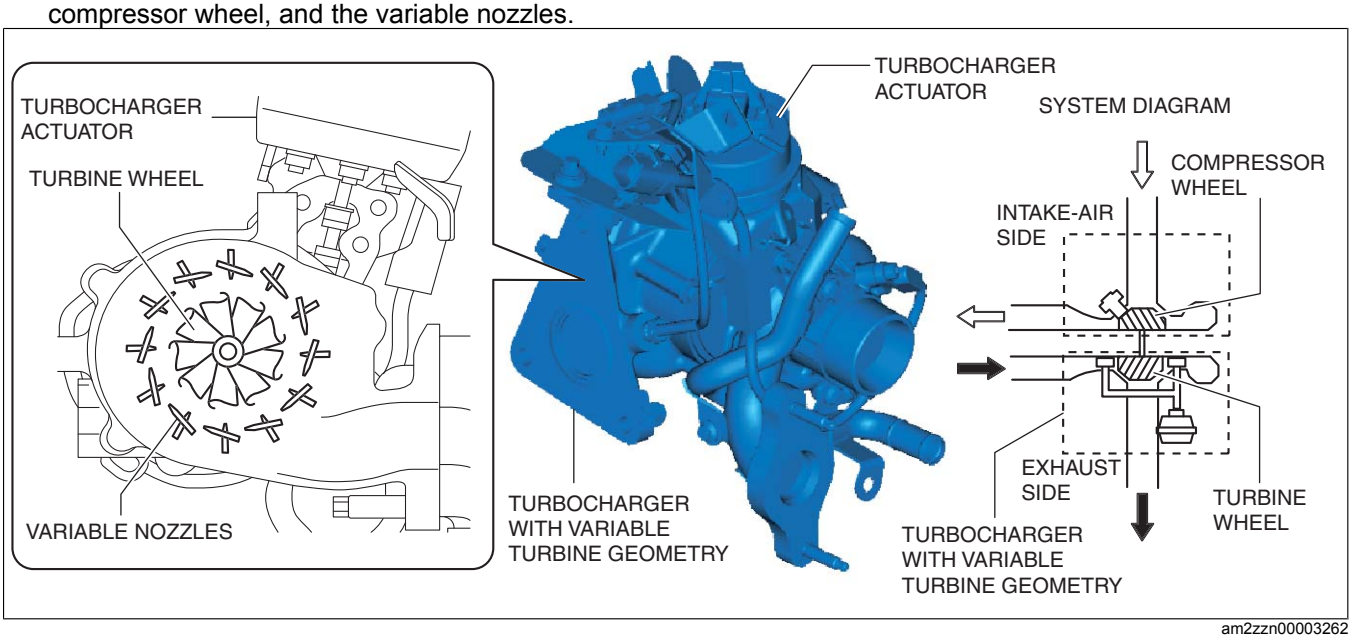
Construction

- The turbocharger with variable turbine geometry is located at the rear of the engine compartment.



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- The turbocharger with variable turbine geometry consists of the turbocharger actuator, turbine wheel, compressor wheel, and the variable nozzles.



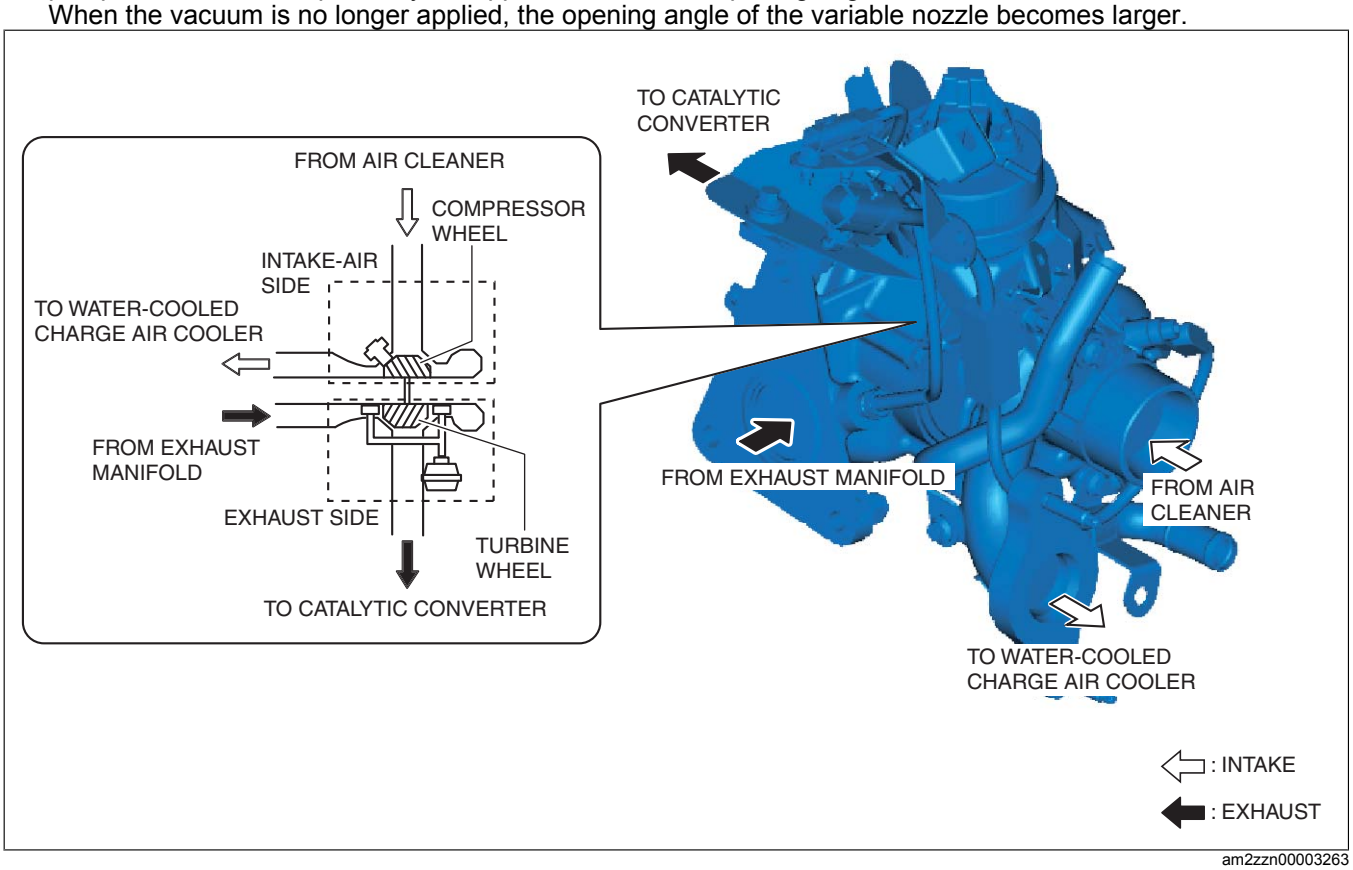
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- To prevent overheating, the turbocharger with variable turbine geometry contains cooling passages through which engine oil lubricates the inside of the turbocharger with variable turbine geometry.

Operation

- With the turbocharger with variable turbine geometry, the turbine wheel and the compressor wheel on the same axis are rotated by the exhaust gas passing through the turbine wheel which charges the intake air.
- The turbocharger with variable turbine geometry is structured such that the total volume of exhaust gas passes through the turbine wheel, and the air charging pressure is performed by the opening and closing of the variable nozzles in the turbocharger with variable turbine geometry.

- The turbocharger actuator rod operates by the adjustment of the turbocharger solenoid valve which occurs from the vacuum generated by the intake manifold and the vacuum in the vacuum chamber generated by the vacuum pump. When the rod is pulled by the applied vacuum, the opening angle of the variable nozzle becomes smaller. When the vacuum is no longer applied, the opening angle of the variable nozzle becomes larger.

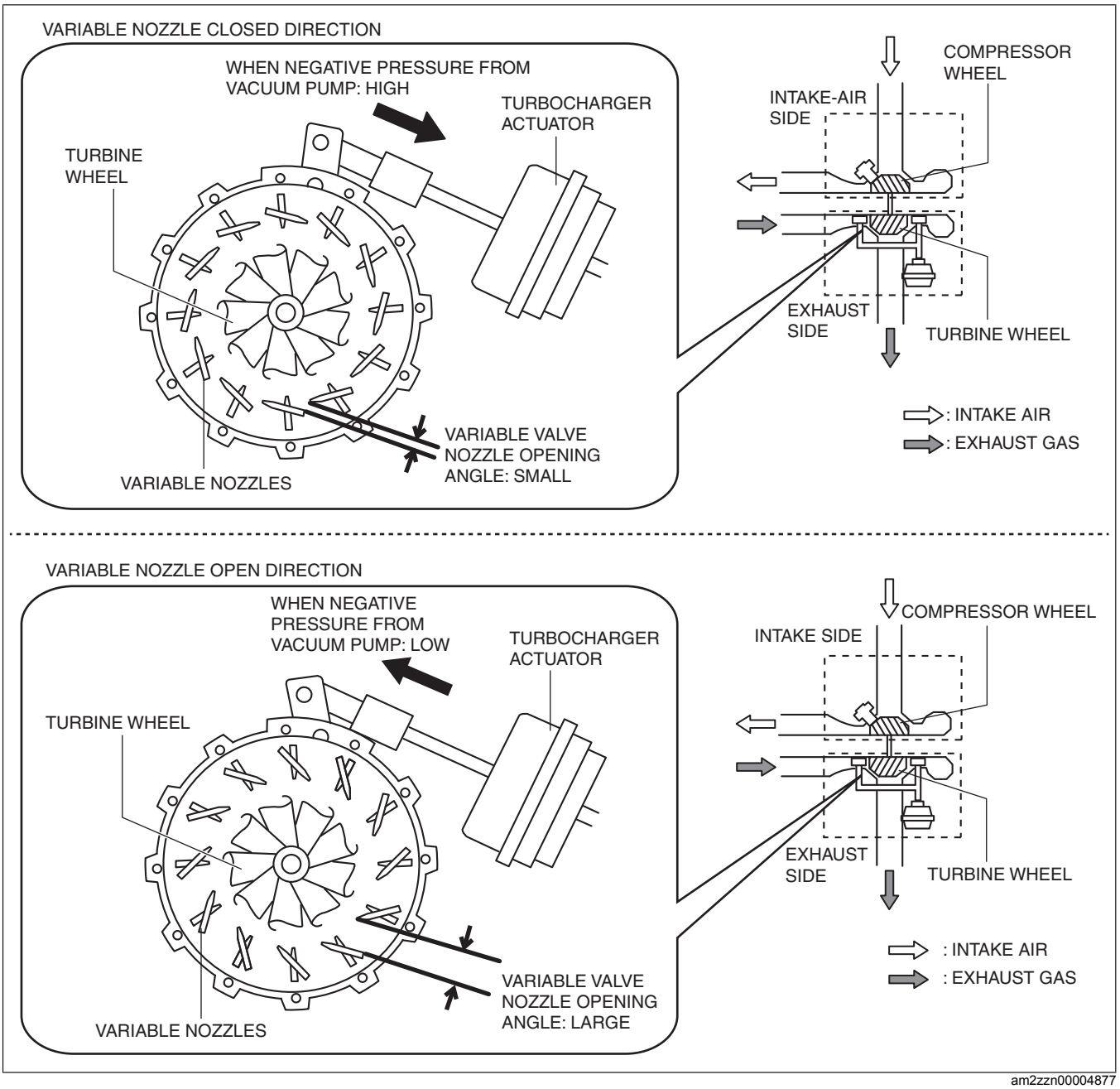


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- The flow speed of the exhaust gas passing through the turbine wheel is varied by the open/closed condition of the variable nozzles. (See AIR CHARGING SYSTEM [SKYACTIV-D 1.5].)

Variable nozzle of the opening angle

- By applying vacuum to the turbocharger actuator, the opening angle of the turbine wheel variable nozzle changes.



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